2025/09 PHY301 Thermodynamics and Statistical Physics

Problems set 1

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Problem 1: Liouville's theorem and entropy [25 marks]

I. In the lecture, we learn that the evolution of the microstates in small time δt is

$$q'_\alpha = q'_\alpha(\{p_\alpha\},\{q_\alpha\}) = q_\alpha + \dot{q}_\alpha(\{p_\alpha\},\{q_\alpha\})\delta t$$
$$p'_\alpha = p'_\alpha(\{p_\alpha\},\{q_\alpha\}) = p_\alpha + \dot{p}_\alpha(\{p_\alpha\},\{q_\alpha\})\delta t.$$

where $\{p_\alpha\} \equiv (p_1,\ldots,p_{3N})$ and $(q_\alpha) \equiv \{q_1,\ldots,q_{3N}\}$. Using the change of variable method and Hamiltonian mechanics to prove that the phase space volume is unchanged up to $\mathcal{O}(\delta t^2)$. Namely, $\prod_{\alpha=1}^{3N} dp'_\alpha dq'_\alpha = \prod_{\alpha=1}^{3N} dp_\alpha dq_\alpha$ where the Jacobian $\frac{\partial(q'_1,\ldots,q'_{3N},p'_1,\ldots,p'_{3N})}{\partial(q_1,\ldots,q_{3N},p_1,\ldots,p_{3N})} = 1+\sum_{\alpha=1}^{3N}(\frac{\partial\dot{q}_\alpha}{\partial q_\alpha}+\frac{\partial\dot{p}_\alpha}{\partial p_\alpha})\delta t+\mathcal{O}(\delta t^2)$.[*Note.* You can argue how the δt^2 arise in the determinant and do not need to evaluate the it exactly.] **[10]**

Solution

The coordinate transformation is

$$dp'_1\ldots dp'_{3N}dq'_1\ldots dq'_{3N} = \frac{\partial(q'_1,\ldots,q'_{3N},p'_1,\ldots,p'_{3N})}{\partial(q_1,\ldots,q_{3N},p_1,\ldots,p_{3N})}dp_1\ldots dp_{3N}dq_1\ldots dq_{3N}$$

The Jacobian is evaluated as the following determinant

$$\frac{\partial(q'_1,\ldots,q'_{3N},p'_1,\ldots,p'_{3N})}{\partial(q_1,\ldots,q_{3N},p_1,\ldots,p_{3N})} = \begin{vmatrix} \frac{\partial p'_1}{\partial p_1} & \cdots & \frac{\partial p'_{3N}}{\partial p_1} & \frac{\partial q'_1}{\partial p_1} & \cdots & \frac{\partial q'_{3N}}{\partial p_1} \\ \frac{\partial p'_1}{\partial p_2} & \cdots & \frac{\partial p'_{3N}}{\partial p_2} & \frac{\partial q'_1}{\partial p_2} & \cdots & \frac{\partial q'_{3N}}{\partial p_2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial p'_1}{\partial p_{3N}} & \cdots & \frac{\partial p'_{3N}}{\partial p_{3N}} & \frac{\partial q'_1}{\partial p_{3N}} & \cdots & \frac{\partial q'_{3N}}{\partial p_{3N}} \\ \frac{\partial p'_1}{\partial q_1} & \cdots & \frac{\partial p'_{3N}}{\partial q_1} & \frac{\partial q'_1}{\partial q_1} & \cdots & \frac{\partial q'_{3N}}{\partial q_1} \\ \frac{\partial p'_1}{\partial q_2} & \cdots & \frac{\partial p'_{3N}}{\partial q_2} & \frac{\partial q'_1}{\partial q_2} & \cdots & \frac{\partial q'_{3N}}{\partial q_2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial p'_1}{\partial q_{3N}} & \cdots & \frac{\partial p'_{3N}}{\partial q_{3N}} & \frac{\partial q'_1}{\partial q_{3N}} & \cdots & \frac{\partial q'_{3N}}{\partial q_{3N}} \end{vmatrix}$$

$$= \begin{vmatrix} (1+\frac{\partial\dot{p}_1}{\partial p_1}\delta t) & \cdots & \frac{\partial\dot{p}_{3N}}{\partial p_1}\delta t & \frac{\partial\dot{q}_1}{\partial p_1}\delta t & \cdots & \frac{\partial\dot{q}_{3N}}{\partial p_1}\delta t \\ \frac{\partial\dot{p}_1}{\partial p_2}\delta t & \cdots & \frac{\partial\dot{p}_{3N}}{\partial p_2}\delta t & \frac{\partial\dot{q}_1}{\partial p_2}\delta t & \cdots & \frac{\partial\dot{q}_{3N}}{\partial p_2}\delta t \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial\dot{p}_1}{\partial p_{3N}}\delta t & \cdots & (1+\frac{\partial\dot{p}_{3N}}{\partial p_{3N}}\delta t) & \frac{\partial\dot{q}_1}{\partial p_{3N}}\delta t & \cdots & \frac{\partial\dot{q}_{3N}}{\partial p_{3N}}\delta t \\ \frac{\partial\dot{p}_1}{\partial q_1}\delta t & \cdots & \frac{\partial\dot{p}_{3N}}{\partial q_1}\delta t & (1+\frac{\partial\dot{q}_1}{\partial q_1}\delta t) & \cdots & \frac{\partial\dot{q}_{3N}}{\partial q_1}\delta t \\ \frac{\partial\dot{p}_1}{\partial q_2}\delta t & \cdots & \frac{\partial\dot{p}_{3N}\delta t}{\partial q_2} & \frac{\partial\dot{q}_1}{\partial q_2}\delta t & \cdots & \frac{\partial\dot{q}_{3N}}{\partial q_2}\delta t \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial\dot{p}_1}{\partial p_{3N}}\delta t & \cdots & \frac{\partial\dot{p}_{3N}}{\partial q_{3N}}\delta t & \frac{\partial\dot{q}_1}{\partial q_{3N}}\delta t & \cdots & (1+\frac{\partial\dot{q}_{3N}}{\partial q_{3N}}\delta t) \end{vmatrix}$$

Solution

Expanding the determinant as follows,

$$\frac{\partial(q'_1,\ldots,q'_{3N},p'_1,\ldots,p'_{3N})}{\partial(q_1,\ldots,q_{3N},p_1,\ldots,p_{3N})} =$$

$$(1+\frac{\partial\dot{p}_2}{\partial p_2}\delta t)\begin{vmatrix} (1+\frac{\partial\dot{p}_2}{\partial p_2}\delta t) & \ldots & \frac{\partial\dot{p}_{3N}}{\partial p_2}\delta t & \frac{\partial\dot{q}_1}{\partial p_2}\delta t & \ldots & \frac{\partial\dot{q}_{3N}}{\partial p_2}\delta t \\ \frac{\partial\dot{p}_2}{\partial p_3}\delta t & \ldots & \frac{\partial\dot{p}_{3N}}{\partial p_3}\delta t & \frac{\partial\dot{q}_1}{\partial p_3}\delta t & \ldots & \frac{\partial\dot{q}_{3N}}{\partial p_3}\delta t \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial\dot{p}_2}{\partial p_{3N}}\delta t & \ldots & (1+\frac{\partial\dot{p}_{3N}}{\partial p_{3N}}\delta t) & \frac{\partial\dot{q}_1}{\partial p_{3N}}\delta t & \ldots & \frac{\partial\dot{q}_{3N}}{\partial p_{3N}}\delta t \\ \frac{\partial\dot{p}_2}{\partial q_1}\delta t & \ldots & \frac{\partial\dot{p}_{3N}}{\partial q_1}\delta t & (1+\frac{\partial\dot{q}_1}{\partial q_1}\delta t) & \ldots & \frac{\partial\dot{q}_{3N}}{\partial q_1}\delta t \\ \frac{\partial\dot{p}_1}{\partial q_2}\delta t & \ldots & \frac{\partial\dot{p}_{3N}\delta t}{\partial q_2} & \frac{\partial\dot{q}_1}{\partial q_2}\delta t & \ldots & \frac{\partial\dot{q}_{3N}}{\partial q_2}\delta t \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial\dot{p}_1}{\partial p_{3N}}\delta t & \ldots & \frac{\partial\dot{p}_{3N}}{\partial q_{3N}}\delta t & \frac{\partial\dot{q}_1}{\partial q_{3N}}\delta t & \ldots & (1+\frac{\partial\dot{q}_{3N}}{\partial q_{3N}}\delta t) \end{vmatrix}$$

$$+\Big(\frac{\partial\dot{p}_2}{\partial p_1}A_2+\cdots+\frac{\partial\dot{p}_{3N}}{\partial p_1}A_{3N}\Big)\delta t+\Big(\frac{\partial\dot{q}_1}{\partial p_1}B_1+\ldots\frac{\partial\dot{q}_{3N}}{\partial p_1}B_{3N}\Big)\delta t \qquad (1)$$

where $A_2\ldots A_{3N}$ and $B_1\ldots B_{3N}$ are the cofactors in Laplace expansion.
By noting that, these cofactors always contain one raw that we can factorize out δt. Namely, $A_2,\ldots,A_{3N},B_1,\ldots,B_{3N}\sim\delta t$. Only the first term in Eq. (1) is in $\mathcal{O}(\delta t)$. Therefore, we keep only the first term. Repeat the same procedure in the Laplace expansion and discarding the $\mathcal{O}(\delta^2)$ terms, we find that

$$\frac{\partial(q'_1,\ldots,q'_{3N},p'_1,\ldots,p'_{3N})}{\partial(q_1,\ldots,q_{3N},p_1,\ldots,p_{3N})} =$$
$$\Big(1+\frac{\partial\dot{p}_1}{p_1}\delta t\Big)\ldots\Big(1+\frac{\partial\dot{p}_{3N}}{p_{3N}}\delta t\Big)\Big(1+\frac{\partial\dot{q}_1}{q_1}\delta t\Big)\ldots\Big(1+\frac{\partial\dot{q}_{3N}}{q_{3N}}\delta t\Big)+\mathcal{O}(\delta t^2)$$
$$=1+\sum_{\alpha=1}^{3N}\Big(\frac{\partial\dot{p}_\alpha}{p_\alpha}+\frac{\partial\dot{q}_\alpha}{q_\alpha}\Big)\delta t+\mathcal{O}(\delta t^2)$$

Using the Hamiltonian mechanics, we arrive at the desirable result $\prod_{\alpha=1}^{3N}dp'_\alpha dq'_\alpha=\prod_{\alpha=1}^{3N}dp_\alpha dq_\alpha$.

II. The normalized ensemble density is a probability in the phase space Γ. This probability has an associated entropy $S(t)=-\int d\Gamma\rho(\Gamma,t)\ln\rho(\Gamma,t)$

(a) Using the method of Lagrange multipliers, find the function $\rho_{\max}(\Gamma)$ that maximizes the functional $S[\rho]$, subject to the constraint of fixed energy, $\langle\mathcal{H}\rangle=\int d\Gamma\rho\mathcal{H}=E$. **[10]**

Solution

To maximizes $S[\rho]$ subject to the energy constraint $\langle\mathcal{H}\rangle - E = 0$, we use the Lagrange multiplier method by maximizing the following function without any constraint as

$$\mathcal{L}[\rho] = S(t) + \alpha(\langle\mathcal{H}\rangle - E = 0)$$

The maximum can be obtained by varying the distribution function ρ as

$$\delta_\rho \mathcal{L}[\rho] = 0 \quad [3]$$

This yields

$$\begin{aligned}
\delta_\rho S(t) + \alpha\delta_\rho\langle\mathcal{H}\rangle &= 0 \\
\int d\Gamma \delta_\rho[\rho(\Gamma,t)\ln\rho(\Gamma,t)] + \alpha\int d\Gamma\delta_\rho[\mathcal{H}\rho(\Gamma,t)] &= 0 \\
\int d\Gamma[\delta_\rho\rho(\Gamma,t)\ln\rho(\Gamma,t) + \rho(\Gamma,t)\delta_\rho\ln\rho(\Gamma,t)] + \alpha\mathcal{H}\delta_\rho\rho(\Gamma,t)] &= 0 \\
\int d\Gamma\delta_\rho\rho(\Gamma,t)[\ln\rho(\Gamma,t) + 1 + \alpha\mathcal{H}] &= 0 \quad [4]
\end{aligned}$$

This implies

$$\begin{aligned}
\ln\rho_{ex}(\Gamma,t) + 1 + \alpha\mathcal{H} &= 0 \\
\rho_{ex}(\Gamma,t) &= C\mathrm{e}^{-\alpha\mathcal{H}}
\end{aligned}$$

where $C = \mathrm{e}^{-1}$. [3]

(b) Show that the solution to part (b) is stationary, that is, $\frac{\partial\rho_{\max}}{\partial t} = 0$. **[5]**

Solution

$$\begin{aligned}
\frac{\partial\rho_{ex}}{\partial t} &= -\{\rho_{ex},\mathcal{H}\} \\
&= -\Big[\frac{\partial\rho_{ex}}{\partial q}\frac{\partial\mathcal{H}}{\partial p} - \frac{\partial\rho_{ex}}{\partial p}\frac{\partial\mathcal{H}}{\partial q}\Big] \\
&= -\Big[-C\alpha\mathrm{e}^{-\alpha\mathcal{H}}\frac{\partial H}{\partial q}\frac{\partial\mathcal{H}}{\partial p} + C\alpha\mathrm{e}^{-\alpha\mathcal{H}}\frac{\partial\mathcal{H}}{\partial p}\frac{\partial\mathcal{H}}{\partial q}\Big] \\
&= 0 \quad [5]
\end{aligned}$$

Problem 2: Maxwell-Boltzmann distribution [25 marks]

Consider the energy of individual atoms in a box with length L in each dimension be $\varepsilon = \frac{1}{2}m(v_x^2 + v_y^2 + v_z^2) + mgz$, where m is the mass, g is the gravity and z is the height ($z \in [0, L]$, one may set $L \to \infty$ for simplicity).

(a) Using Maxwell-Boltzmann theory and letting $\mu = (v_x, v_y, v_z, x, y, z)$ be the states of the particle, at thermal equilibrium with temperature T, what is the probability distribution (ρ) for finding a particle in state μ? **[7]**

Solution

$$\rho(\epsilon) = C\mathrm{e}^{-\varepsilon/(k_B T)} = C\exp\Big[-\frac{\frac{1}{2}mv^2 + mgz}{k_B T}\Big]$$

where C is a normalization constant with

$$\int \rho(\varepsilon)d\mu = 1 \quad \text{with } d\mu = d^3v d^3x$$

Carrying out the above integration, we have

$$C\int \exp\Big[-\frac{m(v_x^2+v_y^2+v_z^2)}{2k_BT}\Big]dv_x dv_y dv_z L^2 \int_0^\infty \exp\Big[-\frac{mgz}{k_BT}\Big]dz = 1$$

$$C\Big\{\int \exp\Big[-\frac{mv_i^2}{2k_BT}\Big]dv_i\Big\}^3 L^2\frac{k_BT}{mg}dz = 1$$

$$C\Big[\sqrt{\frac{2k_BT\pi}{m}}\Big]^3 L^2\frac{k_BT}{mg} = 1$$

Therefore, this yields the probability distribution as

$$\rho(\varepsilon) = \Big(\frac{m}{k_BT}\Big)^{5/2}\frac{g}{\pi^{3/2}L^2}\mathrm{e}^{-\varepsilon/(k_BT)}$$

(b) Prove that the density (n) of the gas at equilibrium $n(z) \propto \exp\Big[-\frac{mgz}{k_BT}\Big]$. **[5]**

Solution

At thermal equilibrium, we can let the particle density equals to the probability density. This leads to

$$n(z) = \int \rho(\mu)d^3v = \frac{mg}{L^2k_BT}\exp\Big[-\frac{mgz}{k_BT}\Big]$$

(c) What is the average energy of a particle $\bar{\varepsilon}$? **[5]**

Solution

$$
\begin{aligned}
\bar{\varepsilon} &= \int \Big[\frac{mv^2}{2} + mgz\Big]\rho(\mu)d\mu \\
&= \Big(\frac{m}{2\pi k_B T}\Big)^{3/2} \int \frac{mv^2}{2} \exp\Big[-\frac{mv^2}{2k_B T}\Big] d^3v \\
&\quad + \frac{mg}{k_B T L^2} \int mgz \exp\Big[-\frac{mgz}{k_B T}\Big] d^3x \\
&= \Big(\frac{m}{2\pi k_B T}\Big)^{3/2} \int_0^\infty \frac{mv^2}{2} \exp\Big[-\frac{mv^2}{2k_B T}\Big] (4\pi v^2) dv \\
&\quad + \frac{mg}{k_B T L^2} \int mgz \exp\Big[-\frac{mgz}{k_B T}\Big] L^2 dz \\
&= \Big(\frac{m}{2\pi k_B T}\Big)^{3/2} \frac{m}{2} \Big(\frac{2k_B T}{m}\Big)^{5/2} 4\pi \int_0^\infty t^4 \mathrm{e}^{-t^2} dt + \frac{mg}{k_B T} mg \Big(\frac{k_B T}{mg}\Big)^2 \int_0^\infty u \mathrm{e}^{-u} du
\end{aligned}
$$

where $t^2 = \frac{m}{2k_B T} v^2$ and $u = \frac{mg}{k_B T} z$. Using Gaussian integral

$$\int_{-\infty}^{\infty} x^{2n} \mathrm{e}^{-\alpha x^2} dx = \frac{d^n}{d\alpha^n} \sqrt{\frac{\pi}{\alpha}}.$$

We obtain

$$\bar{\varepsilon} = \frac{1}{\pi^{3/2}} k_B T (4\pi) \Big(\frac{1}{2}\frac{3}{4}\pi^{1/2}\Big) + k_B T = \frac{5}{2} k_B T$$

(d) What is the energy fluctuation $\bar{\epsilon^2} - \bar{\epsilon}^2$? **[8]**

Solution

Here we separate the average of the square of energy into

$$\bar{\varepsilon^2} = \bar{K^2} + 2\bar{K}\bar{U} + \bar{U^2}$$

where the kinetic energy $K = \frac{1}{2}mv^2$ and the potential energy $U = mgz$. From part (c) we have $\bar{K} = \frac{3}{2}k_B T$ and $\bar{U} = k_B T$ Similarly, the average kinetic energy square of a single particle is

$$
\begin{aligned}
\bar{K^2} &= \Big(\frac{m}{2\pi k_B T}\Big)^{3/2} \int \Big(\frac{1}{2}mv^2\Big)^2 \exp\Big[-\frac{\beta mv^2}{2}\Big] d^3v \\
&= \Big(\frac{m}{2\pi k_B T}\Big)^{3/2} \frac{m^2}{2^2} \int_0^\infty v^4 \exp\Big[-\frac{\beta mv^2}{2}\Big] 4\pi v^2 dv \\
&= \frac{m^2}{2^2}(15/4)(\frac{1}{2}\beta m)^{-2} = \frac{15}{4}/\beta^2 = \frac{15}{4} k_B^2 T^2
\end{aligned}
$$

where $\beta = k_B T$. Similary,

Solution

$$\begin{aligned}\bar{U^2} =& \frac{mg}{k_BTL^2}\int (mgz)^2 \exp\Big[-\frac{mgz}{k_BT}\Big]d^3x = \frac{mg}{k_BT}m^2g^2\int_0^\infty z^2\exp\Big[-\frac{mgz}{k_BT}\Big]dz \\ =& k_B^2T^2\int_0^\infty u^2 \mathrm{e}^{-u}du = 2k_B^2T^2\end{aligned}$$

where we have use $u = \frac{mg}{k_BT}z$. Therefore, we have

$$\bar{\epsilon^2} - \bar{\epsilon}^2 = \Big(\frac{15}{4}k_B^2T^2 + 2\frac{3k_BT}{2}k_BT + 2k_B^2T^2\Big) - \Big(\frac{5}{2}k_BT\Big)^2 = \frac{10}{4}k_B^2T^2 = \frac{5}{2}(k_BT)^2 \quad [8]$$

Problem 3: Relativistic ideal gas [25 marks]

N *indistinguishable* non-interacting relativistic particles move in *one* dimension subject to a Hamiltonian

$$\mathcal{H}(\{p_i, q_i\}) = \sum_{i=1}^{N}[c|p_i| + U(q_i)], \quad \text{with } U(q_i) = \begin{cases} 0, & \text{for } 0 \le q_i \le L \\ \infty, & \text{otherwise} \end{cases}$$

Consider a microcanonical ensemble of total energy E.

(a) Given that

$$\int_{\sum_{i=1}^N c|p_i| \le E} d^N p_i = \frac{2^N (E/c)^N}{N!},$$

and using the result in (a), what is the available phase space volume?

Solution

$$\begin{aligned}\Omega(E) =& \frac{L^N}{N!}\int \delta(\mathcal{H} - E)d^N p \\ \approx& \frac{L^N}{N!}\int_{\sum_{i=1}^N c|p_i| \le E} d^N p \\ =& \frac{(2L)^N (E/c)^N}{(N!)^2} \quad [3]\end{aligned}$$

(b) Compute the entropy $S(E, L, N)$.

Solution

$$\begin{aligned} S(E,L,N) =& k_B \ln \frac{(2L)^N (E/c)^N}{(N!)^2} \\ \approx& k_B \Big[N \ln \frac{2LE}{c} - 2(N \ln N - N) \Big] \\ =& k_B \Big[N \ln \frac{2LE}{c} - N \ln \big(N^2 \mathrm{e}^{-2} \big) \Big] \\ =& N k_B \ln \frac{2\mathrm{e}^2 LE}{N^2 c} \qquad [4] \end{aligned}$$

(c) Calculate the one-dimensional pressure P.(*Hint* compare the first law of thermodynamics $TdS = dE + PdL - \mu dN$ with the differential form of entropy $dS(E,L,N)$. The chemical potential μ is a energy cost for adding and additional particle to the system. The work done by doing that is μdN.)

Solution

$$dS(E,L,N) = d\frac{\partial S}{\partial E} dE + \frac{\partial S}{\partial L} dL + \frac{\partial S}{\partial N} dN$$

Compare to the first law, we obtain

$$\frac{P}{T} = \frac{\partial S}{\partial L}$$

$$\begin{aligned} P =& N k_B T \frac{\partial}{\partial L} \ln \frac{2\mathrm{e}^2 LE}{N^2 c} \\ P =& \frac{N k_B T}{L} \qquad [5] \end{aligned}$$

(d) Obtain the heat capacities C_L with constant volume and C_P with constant pressure.(*Hint*. For C_p, you have to account for the work done to the external pressure.)

Solution

$$\frac{1}{T} = \frac{\partial S}{\partial E}$$

$$\begin{aligned} 1/T =& N k_B \frac{\partial}{\partial E} \ln \frac{2\mathrm{e}^2 LE}{N^2 c} \\ \frac{1}{N k_B T} =& \frac{1}{E} \end{aligned}$$

This gives

$$E = N k_B T$$

Solution

Therefore, the heat capacity with constant volume is

$$\left(\frac{\partial E}{\partial T}\right)_{L,N} = Nk_B$$

For constant pressure, the heat capacity is

$$C_P = \frac{dQ}{dT} = \frac{dE + dW}{dT}$$

where we have used the 1st law. From the equation of state, the work done by a constant pressure is

$$dW = PdL = Nk_B dT$$

Therefore, we obtain

$$C_P = 2Nk_B \quad [5]$$

(e) What is the probability $p(p_1)$ of finding a paritcle with momentum p_1?

Solution

First, we count the total number of states (Ω') with fix $|p_1|$. This is given by

$$\begin{aligned}
\bar{\Omega} =& \frac{L^N}{N!}\int \delta(\mathcal{H} - E) dp_2 \dots dp_N \\
\approx& \frac{L^N}{N!}\int_{\sum_{i=2}^N c|p_i| \le E - |p_1|c} d^{N-1}p \\
=& \frac{L}{N}\Big[\frac{L^{N-1}}{(N-1)!}\int_{\sum_{i=2}^N c|p_i| \le E - |p_1|c} d^{N-1}p\Big] \\
=& \frac{L}{N}\Omega(E - |p_1|c, N-1)
\end{aligned}$$

$$\begin{aligned}
p(p_1) =& \frac{\bar{\Omega}(E,N)}{\Omega(E,N)} \\
=& \frac{L/N\Omega(E - |p_1|c, N-1)}{\Omega(E,N)} \\
=& \frac{L(2L)^{N-1}((E - |p_1|c)/c)^{N-1}}{(2L)^N (E/c)^N}\frac{(N!)^2}{N!(N-1)!} \\
=& \frac{((E - |p_1|c)/c)^{N-1}}{2(E/c)^N}N \\
=& \frac{(1 - |p_1 c|/E)^{N-1}}{2(E/c)}N
\end{aligned}$$

Solution

Using $E = Nk_BT$, we have

$$\begin{aligned} p(p_1) =&(1 - \frac{|p_1|c/k_BT)}{N})^{N-1} \frac{1}{2(Nk_BT/c)} N \\ =&(1 - \frac{|p_1|c/k_BT)}{N})^{N-1} \frac{c}{2k_BT} \end{aligned}$$

Taking $N \to \infty$, we arrive at

$$p(p_1) = \frac{c}{2k_BT} \mathrm{e}^{-|p_1|c/k_BT}$$

[5]

Problem 4: Equidistant energy levels system in microcanonical ensemble [25 marks]

In the lecture, we discuss two-level system. Here, we consider N independent quantum oscillators with infinite number of energy level. The Hamiltonian is

$$\mathcal{H}(\{n_i\}) = \sum_{i=1}^{N} \hbar\omega \left(n_i + \frac{1}{2} \right),$$

where $n_i = 0, 1, 2, \ldots$ is energy level for the ith oscillator. In this system, the energy difference between adjacent levels is a constant $\hbar\omega$ [Stirling approximation $\ln N! \approx N \ln N - N$]

(a) Using microcanonical ensemble, calculate the entropy S, as a function of total energy E. [*Hint.* The total number of microstates $\Omega(E)$ can be regarded as the number of ways by permuting $M = \sum_i n_i$ balls and $N - 1$ partition walls together along a line.]

Solution

We the total number of microstate are

$$\Omega = \frac{(M + N - 1)!}{M!(N - 1)!}$$

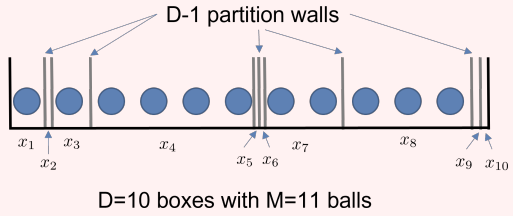


Figure 1: where $D = N$ is the number of boxes.

Solution

The total energy is

$$E = \hbar\omega\Big(\sum_{i=1}^{N} n_i + \frac{N}{2}\Big) = \hbar\omega\Big(M + \frac{N}{2}\Big)$$

Therefore,

$$M = \frac{E}{\hbar\omega} - \frac{N}{2}$$

Then, we have

$$\Omega(E,N) = \frac{(N+M-1)!}{M!(N-1)!} = \frac{(\frac{E}{\hbar\omega} + \frac{N}{2} - 1)!}{(\frac{E}{\hbar\omega} - \frac{N}{2})!(N-1)!}$$

The entropy is

$$S = k_B \ln\Omega = \Big(\frac{E}{\hbar\omega} + \frac{N}{2}\Big)\ln\Big(\frac{E}{\hbar\omega} + \frac{N}{2}\Big) - \Big(\frac{E}{\hbar\omega} - \frac{N}{2}\Big)\ln\Big(\frac{E}{\hbar\omega} - \frac{N}{2}\Big) - N\ln N$$

[6]

(b) Calculate the energy E, and the heat capacity C, as functions of temperature T, and N.

Solution

$$\frac{1}{T} = \frac{\partial S}{\partial E} = \frac{k_B}{\hbar\omega}\ln\frac{E + N\hbar\omega/2}{E - N\hbar\omega/2}$$

We express E in term of T

$$\frac{E + N\hbar\omega/2}{E - N\hbar\omega/2} = \exp\Big[\frac{\hbar\omega}{k_B T}\Big]$$

This leads to

$$E = \frac{N\hbar\omega}{2}\frac{\exp\Big[\frac{\hbar\omega}{k_B T}\Big] + 1}{\exp\Big[\frac{\hbar\omega}{k_B T}\Big] - 1}$$

The Heat capacity is

$$C = \frac{\partial E}{\partial T} = N k_B\Big(\frac{\hbar\omega}{k_B T}\Big)^2 \frac{\exp\Big[\frac{\hbar\omega}{k_B T}\Big]}{[\exp\Big[\frac{\hbar\omega}{k_B T}\Big] - 1]^2}$$

[6]

(c) Find the probability $p(n)$ that a particular oscillator is in its nth quantum level.

Solution

$$
\begin{aligned}
p(n_1) = \sum_{n_2,n_3,\ldots,n_N} p(n_1,\ldots n_N) &= \frac{\Omega(E-(n_1+\frac{1}{2})\hbar\omega, N-1)}{\Omega(E,N)} \\
&= \frac{\frac{(\frac{E}{\hbar\omega}-n_1-\frac{1}{2}+\frac{N-1}{2}-1)!}{(\frac{E}{\hbar\omega}-n_1-\frac{1}{2}-\frac{N-1}{2})!(N-2)!}}{\frac{(\frac{E}{\hbar\omega}+\frac{N}{2}-1)!}{(\frac{E}{\hbar\omega}-\frac{N}{2})!(N-1)!}} \\
&= \frac{\frac{(\frac{E}{\hbar\omega}-n_1+\frac{N}{2}-2)!}{(\frac{E}{\hbar\omega}-n_1-\frac{N}{2})!(N-2)!}}{\frac{(\frac{E}{\hbar\omega}+\frac{N}{2}-1)!}{(\frac{E}{\hbar\omega}-\frac{N}{2})!(N-1)!}}
\end{aligned}
$$

Using Stirling approximation,

$$N! = \mathrm{e}^{N\ln N - N}$$

This gives

$$
\begin{aligned}
p(n_1) =& (N-1)\exp\Big[(\frac{E}{\hbar\omega}-n_1+\frac{N}{2}-2)\ln\Big(\frac{E}{\hbar\omega}-n_1+\frac{N}{2}-2\Big) \\
&-(\frac{E}{\hbar\omega}-n_1-\frac{N}{2})\ln\Big(\frac{E}{\hbar\omega}-n_1-\frac{N}{2}\Big) \\
&-(\frac{E}{\hbar\omega}+\frac{N}{2}-)\ln\Big(\frac{E}{\hbar\omega}+\frac{N}{2}-1\Big)+(\frac{E}{\hbar\omega}-\frac{N}{2})\ln\Big(\frac{E}{\hbar\omega}-\frac{N}{2}\Big)\Big] \\
=& (N-1)\exp\Big[(\frac{E}{\hbar\omega}-n_1+\frac{N}{2}-2)\ln\Big(\frac{E}{\hbar\omega}+\frac{N}{2}-1\Big) \\
&+(\frac{E}{\hbar\omega}-n_1+\frac{N}{2}-2)\ln\Big(1-\frac{n_1+1}{\frac{E}{\hbar\omega}+\frac{N}{2}-1}\Big) \\
&-(\frac{E}{\hbar\omega}-n_1-\frac{N}{2})\ln\Big(\frac{E}{\hbar\omega}-\frac{N}{2}\Big) \\
&-(\frac{E}{\hbar\omega}-n_1-\frac{N}{2})\ln\Big(1-\frac{n_1+1}{\frac{E}{\hbar\omega}-\frac{N}{2}}\Big) \\
&-(\frac{E}{\hbar\omega}+\frac{N}{2}-1)\ln\Big(\frac{E}{\hbar\omega}+\frac{N}{2}-1\Big)+(\frac{E}{\hbar\omega}-\frac{N}{2})\ln\Big(\frac{E}{\hbar\omega}-\frac{N}{2}\Big)\Big]
\end{aligned}
$$

Solution

$$\begin{aligned} p(n_1) =&(N-1)\exp\Big[-(n_1+1)\ln\left(\frac{E}{\hbar\omega}+\frac{N}{2}-1\right) \\ &+(\frac{E}{\hbar\omega}-n_1+\frac{N}{2}-2)\ln\left(1-\frac{n_1+1}{\frac{E}{\hbar\omega}+\frac{N}{2}-1}\right) \\ &+n_1\ln\left(\frac{E}{\hbar\omega}-\frac{N}{2}\right)\Big] \\ &-(\frac{E}{\hbar\omega}-n_1-\frac{N}{2}-1)\ln\left(1-\frac{n_1+1}{\frac{E}{\hbar\omega}-\frac{N}{2}}\right) \end{aligned}$$

Since $E \gg n_1\hbar\omega$ and $N \gg 1$, we then obtain

$$p(n_1) \approx N\exp\Big[-(n_1+1)\ln\left(\frac{E}{\hbar\omega}+\frac{N}{2}\right)+n_1\ln\left(\frac{E}{\hbar\omega}-\frac{N}{2}\right)\Big]$$

We the temperature expression in part (b), we have

$$\begin{aligned} p(n_1) \approx& \frac{N}{\frac{E}{\hbar\omega}+\frac{N}{2}}\left(\frac{\frac{E}{\hbar\omega}-\frac{N}{2}}{\frac{E}{\hbar\omega}+\frac{N}{2}}\right)^{n_1} \\ =&[1+\mathrm{e}^{-\hbar\omega/(k_BT)}]\mathrm{e}^{-n_1\hbar\omega/(k_BT)} \end{aligned}$$

where we have used the result in part (b) [8]